TargetScan

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1. What is it?

Who developed it?

TargetScan was developed by the **Bartel lab** (Whitehead Institute for Biomedical Research / Howard Hughes Medical Institute). Key contributors across releases include David P. Bartel, Benjamin P. Lewis, Alexander Grimson, Robin C. Friedman, Vikram Agarwal, Sean E. McGeary, and colleagues.

When was it introduced?

- **2005** — seed-based target prediction (Lewis et al., 2005)
- **2007** — context scores and site-type efficacy (Grimson et al., 2007)
- **2009** — preferential conservation / PCT (Friedman et al., 2009)
- **2011** — context+ with TA and SPS (García et al., 2011)
- **2015** — Context++ and TargetScan 7.0 (Agarwal et al., 2015)
- **2020** — TargetScan 8.0 default ranking via biochemical model (McGeary et al., 2019)

Current / relevant version

TargetScan 8.0 (January 2020 ranking update). Mammalian portals:

- Human: TargetScanHuman 8.0
- Mouse: TargetScanMouse 8.0

Release 8 uses the **same 3′ UTR profiles and miRNA families as Release 7**; the main mammalian change is the **default ranking method** (McGeary et al., 2019), with Context++/PCT retained as options.

What problem does it solve?

TargetScan predicts which mRNAs are regulated by which miRNAs by scanning 3′ UTRs (and some ORF regions) for **canonical seed-matched sites**, scoring site efficacy and evolutionary conservation, and ranking gene–miRNA pairs for hypothesis generation. It separates **predicted biochemical repression** from **evolutionary conservation evidence**, allowing users to prioritize either cellular response or biological consequence.

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2. How does it work?

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miRNA families (shared seed) → scan representative 3′ UTR profile
→ classify site type (8mer / 7mer-m8 / 7mer-A1 / 6mer; optional 3′ compensatory)
→ compute conserved branch length & PCT (conservation track)
→ compute Context++ per site (efficacy track, linear model)
→ aggregate to gene–miRNA (total context++, CWCS, aggregate PCT)
→ rank targets (8.0 default: biochemical predicted occupancy; options: CWCS or aggregate PCT)
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Step-by-step process

1. **Curate miRNA families** — group mature miRNAs by identical seed (positions 2–8); classify family conservation (broadly conserved, conserved, poorly conserved).
2. **Define 3′ UTR profiles** — Gencode annotations extended with RefSeq, RNA-seq, and 3P-seq tags; **representative transcript** = variant with most 3P-seq tags (TargetScan 7–8).

3. **Identify candidate sites** — search representative 3′ UTRs for perfect seed matches; orthologous UTRs from UCSC multiz alignments enable conservation analysis.
4. **Score each site** — branch length → conserved/nonconserved; PCT (probability of conserved targeting); Context++ (14-feature efficacy model).
5. **Aggregate per gene–miRNA family** — resolve overlapping sites; compute total context++ score, cumulative weighted context++ score (CWCS), and aggregate PCT.
6. **Rank predictions** — TargetScan 8.0 defaults to **predicted occupancy** from the McGeary biochemical model; users can switch to CWCS or aggregate PCT ranking.

Candidate site identification

- **Canonical sites:** Watson–Crick pairing to miRNA seed (positions 2–7 or 2–8), classified as 8mer, 7mer-m8, 7mer-A1, or 6mer (see Section 8).
- **3′ compensatory sites:** imperfect seed match compensated by conserved 3′ pairing (Friedman et al., 2009); negative site-type codes in download files.
- **Centered sites:** included in TargetScan 7; **removed in TargetScan 8.0** (McGeary et al., 2019 showed they are not functional for most miRNAs).
- **6mer sites:** included in human TargetScan 7+ predictions; always classified **nonconserved** in TargetScanHuman 7+.
- **Conservation filter:** site conservation uses **conserved branch length** thresholds that differ by site type and release (FAQs).

Scoring and ranking

TargetScan provides **three independent ranking axes** for mammalian predictions:

Rank by	Level	Evidence	Default in
Predicted occupancy (biochemical model)	gene–miRNA	efficacy	**TargetScan 8.0 web**
Cumulative weighted context++ score (CWCS)	gene–miRNA	efficacy	TargetScan 7.x web; 8.0 option
Aggregate PCT	gene–miRNA	conservation	Optional in 7.x and 8.0

TargetScan explicitly recommends choosing thresholds at the **target (gene–miRNA) level** rather than individual sites, because multiple weak sites can sum to strong repression (FAQs).

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3. What biological/computational features does it use?

Feature	Used?	Source / notes
Seed matching	✅	Lewis et al., 2005; positions 2–7/8
8mer / 7mer-m8 / 7mer-A1 / 6mer site types	✅	Lewis 2005; Friedman 2009; FAQs site-type column
Evolutionary conservation	✅	Branch length + PCT (Friedman et al., 2009)
Binding / free energy	Partial	SPS (seed-pairing stability) in Context++; McGeary model uses measured/predicted K_D, not ΔG alone

RNA / site accessibility	✅	SA feature in Context++ (RNAplfold; Agarwal et al., 2015)
Local AU content	✅	Context++ feature (Grimson et al., 2007)
Target-site abundance	✅	TA feature (García et al., 2011)
Position within transcript	✅	Minimum distance from 3′ UTR ends (Context++)
3′ compensatory pairing	✅	Supplementary pairing feature; separate 3′ compensatory site class
G:U pairing	Partial	sRNA/site position-8 identity features in Context++; not primary seed rule
Machine learning	✅ (8.0 default)	CNN predicts K_D for McGeary biochemical model; Context++ itself is **multiple linear regression**, not ML
Other sequence/context features	✅	3′ UTR length, ORF length, ORF 8mer count, offset-6mer count, 3P-seq AIR weighting

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4. What does its score mean?

Question	Answer
Score name(s)	Context++, total context++ score, cumulative weighted context++ score (CWCS), PCT, predicted occupancy (biochemical model), predicted relative K_D
Numerical range	Context++: site-type-bounded (8mer ≤ −0.03, 7mer-m8 ≤ −0.02, 7mer-A1 ≤ −0.01, 6mer ≤ 0). CWCS: $\log_2$ scale (unbounded upper; higher = stronger). PCT: 0–1. Predicted occupancy: fraction of AGO–miRNA complex on target sites (0–1 range).
Higher or lower = stronger?	Context++: **more negative** = stronger predicted site repression. CWCS / predicted occupancy: **higher** = stronger predicted gene-level repression. PCT: **higher** = stronger conservation evidence.
Score type	Context++: linear regression contribution (log-scale repression term). CWCS: $\log_2$ cumulative repression. PCT: probability. Biochemical: predicted occupancy fraction.
Absolute interpretation across genes?	Context++ and CWCS are calibrated on experimental compendia but **not guaranteed comparable across miRNA families** without percentile ranks. PCT is comparable as probability. Occupancy depends on assumed miRNA concentration scenario.
Already normalized?	Context++ features are trimmed-normalized to (0,1) before regression (Agarwal et al., 2015). Percentile columns provided in downloads. PCT is inherently normalized 0–1.
Recommended cutoff / threshold	**No universal cutoff** — FAQs defer to user sensitivity/specificity trade-offs; thresholds differ by miRNA family and biological question.
Site scores aggregated to gene–miRNA score?	Yes — total context++ (sum of non-overlapping sites), CWCS (AIR-weighted marginal repression), aggregate PCT (gene-level conservation rollup).

4b. Score inventory

Score name	Level	Evidence type	Range	Direction	Notes
Context++	site	efficacy	Site-type bounded; 8mer ≤ −0.03	More negative = stronger	Sum of 14 feature contributions; separate linear model per site type
Total context++ score	gene–miRNA	efficacy	Unbounded (sum of site scores)	More negative sum = stronger	Sum over **non-overlapping** sites maximizing total
Weighted Context++ (CWCS)	gene–miRNA	efficacy	$\log_2$; typically ~0 to ~5+	Higher = stronger	$CWCS = \log_2(1 - C_n)$ from AIR-weighted marginal repression
PCT	site; aggregate at gene–miRNA	conservation	0–1	Higher = stronger	Probability of preferential conservation; **separate ranking axis from Context++**
Biochemical model score (predicted occupancy)	gene–miRNA	efficacy	~0–1 (fraction)	Higher = stronger	**TargetScan 8.0 default ranking**; McGeary CNN + biochemical model; three concentration scenarios in downloads

Score interpretation notes

- **Context++** = predicted targeting efficacy at a single site (biochemical response to miRNA).
- **PCT** = probability that a site is conserved **because** it is targeted (evolutionary evidence of biological consequence).
- **PCT is also one of the 14 Context++ features**, but the standalone PCT score and aggregate PCT ranking remain **conservation readouts** — do not treat Context++ and PCT as interchangeable outputs.
- **CWCS vs biochemical occupancy:** both predict efficacy; biochemical model (8.0 default) outperformed retrained Context++ in McGeary et al., 2019 cross-validation, but uses a different mathematical framework (K_D + occupancy → repression).
- **No recommended cutoffs** — use target-level thresholds based on your sensitivity/specificity needs (FAQs).

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5. Advantages

- Extensively calibrated against large experimental compendia (74 transfection datasets for Context++; AGO-RBNS + transfection data for biochemical model).
- Separates **efficacy** (Context++/CWCS/biochemical) from **conservation** (PCT), avoiding conflation of biochemical and evolutionary evidence.
- Transparent **site-level output** with site type, context++ score, branch length, and PCT per site.

- Long publication lineage with downloadable parameters, Perl scripts, and bulk files for reproducible offline analysis.
- Representative-transcript and 3P-seq AIR weighting reduces isoform confounding (Nam et al., 2014; Agarwal et al., 2015).
- TargetScan 8.0 retains legacy Context++/PCT ranking as explicit options while improving default efficacy ranking.

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6. Disadvantages

- **3′ UTR–centric** — standard gene pages emphasize 3′ UTR sites; ORF sites exist but are secondary.
- **Representative transcript choice** affects which sites appear; genes with complex isoforms may be under- or over-represented.
- **Multiple incompatible ranking scores** (occupancy vs CWCS vs aggregate PCT) — 8.0 default differs from 7.x, creating integration hazards across tools and releases.
- **No universal score threshold** — users must define cutoffs per study.
- **6mer-only targets** are scored but not displayed on gene pages; targets with only 6mer sites are not listed (Agarwal et al., 2015).
- **McGeary CNN K_D predictions** extrapolate from six experimentally measured miRNAs — accuracy may drop for miRNAs with atypical seed-binding patterns.
- Poorly conserved miRNA families include misannotated small RNAs (FAQs).

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7. When would you use it?

Use TargetScan when you need **seed-based mammalian target predictions** with explicit separation of predicted repression and evolutionary conservation — for example, ranking PTEN targets of miR-21 by CWCS while flagging high aggregate PCT sites as evolutionarily supported.

- Choose **predicted occupancy** (8.0 default) for best-validated cellular repression ranking.
- Choose **CWCS** for compatibility with TargetScan 7 literature and linear Context++ framework.
- Choose **aggregate PCT** when biological consequence / conserved targeting matters more than biochemical potency (especially for broadly conserved miRNA families).
- Always confirm **miRNA and mRNA co-expression** in your cell type — TargetScan does not filter by expression (FAQs).

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8. Method-specific deep dive

- ☑ **Seed matching** — Canonical sites require Watson–Crick pairing to miRNA positions 2–7 (6mer, 7mer-A1) or 2–8 (7mer-m8, 8mer). miRNA families grouped by identical seed. (Lewis et al., 2005; FAQs)
- ☑ **8mer sites** — Exact match to positions 2–8 of miRNA **plus** an adenosine at target position 1 (immediately 3′ of the seed match). Strongest canonical site type. Download code: `3`. (FAQs; Lewis et al., 2005; Friedman et al., 2009)
- ☑ **7mer-m8 sites** — Exact match to positions 2–8 (seed + position 8). Second-strongest canonical type. Download code: `2`. (FAQs; Friedman et al., 2009)
- ☑ **7mer-A1 sites** — Exact match to positions 2–7 plus adenosine opposite miRNA position 1. Download code: `1`. (FAQs; Lewis et al., 2005)
- ☑ **6mer sites** — Exact match to positions 2–7 only. Weakest canonical type; always nonconserved in TargetScanHuman 7+. Download code: `4`. (FAQs; Friedman et al., 2009; Agarwal et al., 2015)
- ☑ **Relative site efficacy** — Experimental hierarchy: 8mer > 7mer-m8 > 7mer-A1 > 6mer (Friedman et al., 2009; Grimson et al., 2007 site-type weights 0.31 / 0.161 / 0.099 for fly sorting). Context++ uses learned site-

type intercepts per type (Agarwal et al., 2015 Figure 4).

- **Context++ score** — For each site, CS = sum of 14 feature contributions from a **site-type-specific multiple linear regression** trained on 74 sRNA transfection datasets. Features: site type, supplementary pairing, local AU, minimum distance, sRNA1A/C/G, sRNA8A/C/G, site8A/C/G, 3′ UTR length, SA (site accessibility), ORF length, ORF 8mer count, 3′ UTR offset 6mer count, TA, SPS, PCT. Predicted site repression = `1 – 2^CS`. More negative CS → stronger repression. (Agarwal et al., 2015; context_score.html)
- **Weighted Context++ score (CWCS)** — Iterative marginal repression accounting for 3P-seq AIR (isoform abundance) and downstream-site depletion: $C_i = C(i-1) + (1 - 2^{CS_i})(AIR_i - C(i-1))$, with $C_0 = 0$; $CWCS = \log_2(1 - C_n)$ where C_n is cumulative repression at the most proximal site. Higher CWCS = greater predicted repression. (Agarwal et al., 2015; context_score_totals.html)
- **Total context++ score** — After removing overlapping sites (keeping set that **maximizes** total context score), sum Context++ scores for non-overlapping sites to the representative miRNA. All overlapping sites still shown on gene detail pages. (context_score_totals.html; Agarwal et al., 2015)
- **PCT** — Probability that site conservation exceeds background **because of miRNA targeting**, estimated relative to 50 control k-mers matched for length and dinucleotide conservation rates, using phylogenetic branch lengths binned by UTR conservation (Friedman et al., 2009; updated in Agarwal et al., 2015). **Aggregate PCT** rolls up to gene–miRNA for conservation-based ranking. Contrast: PCT measures evolutionary evidence; Context++ measures predicted biochemical efficacy. PCT as a Context++ feature adds conservation signal to efficacy prediction but does not replace standalone PCT ranking.
- **Evolutionary conservation model** — Conserved branch length (BLS) thresholds vary by site type and release (e.g., TargetScanHuman 7–8: 8mer ≥ 1.8, 7mer-m8 ≥ 2.6, 7mer-A1 ≥ 3.6). miRNA family conservation: broadly conserved (to zebrafish), conserved (placental mammals), poorly conserved. Site and family conservation are **independent** dimensions. (FAQs Release 8)
- **Ranking — web vs download** — **Web (8.0):** default sort by biochemical predicted occupancy; options for CWCS and aggregate PCT. **Downloads:** `Summary_Counts` files provide Total context++ score, Cumulative weighted context++ score, Aggregate PCT, and Predicted occupancy (low/high/transfected miRNA). Site-level files provide context++ score, percentiles, weighted context++ score, predicted relative K_D. BED files use context++ score percentile as score. (TargetScan 8.0 data download page)
- **Version changes 7.x → 8.0** — (1) Default mammalian ranking → McGeary biochemical model + CNN. (2) Centered sites removed. (3) Same 3′ UTR profiles and miRNA families as Release 7. (4) Non-conserved target tables more accessible. (5) Context++/PCT ranking preserved as options. (TargetScan 8.0 help page; vert_80 home page)
- **Score interpretation guide** — For highly conserved miRNA families, choose aggregate PCT (biological consequence / conserved targeting) **or** CWCS/context scores (biochemical efficacy, works for non-conserved miRNAs too). No single best cutoff. Consider co-expression. (FAQs Q11–12)
- **McGeary 2019 biochemical model** — AGO-RBNS measured relative K_D for six miRNAs across ≤12-nt sites. Biochemical model predicts AGO–miRNA occupancy on each mRNA from site K_D values and miRNA concentration, then predicts repression. CNN extends K_D prediction to all miRNAs from sequence. TargetScan 8.0 uses this as **default ranking** via predicted occupancy columns. Outperformed retrained Context++ in held-out miRNA cross-validation (McGeary et al., 2019).

Site-level → gene–miRNA aggregation

1. **Filter sites** — For each miRNA family, retain 8mer, 7mer-m8, 7mer-A1, 6mer sites; remove overlaps (distal-to-proximal numbering for CWCS).
2. **Representative miRNA** — Compute CWCS for each family member; choose member with greatest predicted repression (ties: lowest miRNA name/number).
3. **Representative transcript** — UTR profile with most 3P-seq tags represents the gene.
4. **Outputs per gene–miRNA family:**

- **Total context++ score** — sum of Context++ for non-overlapping sites to representative miRNA.
- **CWCS** — AIR-weighted cumulative repression (formula above).
- **Aggregate PCT** — conservation rollup for ranking (Friedman et al., 2009 method).
- **Predicted occupancy** — biochemical model fraction of AGO–miRNA on target sites (8.0 default rank).

Poorly conserved sites receive the **same weight as conserved sites** in CWCS calculation (context_score_totals.html).

Sources

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- García DM, Baek D, Shin C, Bell GW, Grimson A, Bartel DP (2011). Weak seed-pairing stability and high target-site abundance decrease the proficiency of lsy-6 and other miRNAs. *Nat Struct Mol Biol* 18:1139–1146.
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- McGeary SE, Lin KS, Shi CY, Pham T, Bisaria N, Kelley GM, Bartel DP (2019). The biochemical basis of microRNA targeting efficacy. *Science* 366(6472):eaav1741.
- TargetScan FAQs Release 7+: https://www.targetscan.org/faqs.Release_7.html
- TargetScan FAQs Release 8: https://www.targetscan.org/faqs.Release_8.html
- TargetScan context++ documentation: https://www.targetscan.org/vert_71/docs/context_score.html , https://www.targetscan.org/vert_71/docs/context_score_totals.html
- TargetScanHuman 8.0: https://www.targetscan.org/vert_80/
- TargetScan 8.0 data download (field descriptions): https://www.targetscan.org/cgi-bin/targetscan/data_download.vert80.cgi
- TargetScan 8.0 changelog: https://www.targetscan.org/vert_80/docs/help.html

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Integration notes (for consensus project)

Recommended R-tool export (this project):

Column	Source field	Role
`cwcs`	`Cumulative weighted context++ score`	Primary efficacy rank (stable across 7.x/8.0 literature)
`aggregate_pct`	`Aggregate PCT`	Conservation annotation overlay — **not averaged with efficacy**
`biochem_occupancy`	`Predicted occupancy (transfected miRNA)`	Optional 8.0-default efficacy rank; document which concentration scenario was chosen

TargetScan 8.0 bulk files to use (`scripts/download_targetscan.R`):

- **Gene–miRNA pairs:** `Summary_Counts.all_predictions.txt.zip` (full) or `Summary_Counts.default_predictions.txt.zip` (conserved sites of conserved families only)
- **Site-level Context++:** `Conserved_Site_Context_Scores.txt.zip` + `Nonconserved_Site_Context_Scores.txt.zip`
- **Site positions + PCT:** `Conserved_Family_Info.txt.zip` + `Nonconserved_Family_Info.txt.zip`

Score selection guidance:

- Default to **CWCS** for cross-tool comparison with DIANA/miRDB unless the analysis explicitly requires TargetScan 8.0 web-default behavior (then use predicted occupancy).
- Attach **aggregate PCT** as a separate conservation tier, not merged into efficacy.
- **Do not** naively average Context++, PCT, CWCS, biochemical occupancy, or TargetScan scores with DIANA/miRDB/mirDIP scores — different evidence types and scales.
- When mirDIP is used, remember it may already include TargetScan-derived evidence (correlated, not independent).

PTEN example (interpretation): The same PTEN–miRNA pair may rank high by CWCS (strong 8mer sites, favorable context) but lower by aggregate PCT (sites not preferentially conserved), explaining disagreement with conservation-heavy tools or with PCT-sorted TargetScan views.